Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Non-metals combine with oxygen to form usually

(A) Basic oxides

(B) Neutral oxides

(C) Acidic oxides

(D) Amphoteric oxides

Q.2

The product A in the following reaction:
$2KMnO_4 \longrightarrow A + KMnO_2 + O_2$ is

(A) $K_2Mn_2O_7$

(B) K_2MnO_4

(C) K_2O

(D) K_2O_2

Q.3

On heating ozone, its volume

(A) Increases to 1.5 times

(B) Decreases to half

(C) Remain unchanged

(D) Becomes double

Q.4

The correct order of $O-O$ bond length in O_2, H_2O_2 and O_3

(A) $O_3 > H_2O_2 > O_2$

(B) $O_2 > H_2O_2 > O_3$

(C) $O_2 > O_3 > H_2O_2$

(D) $H_2O_2 > O_3 > O_2$

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Q.5

Tailing of mercury test can be used for which of the following gas?

(A) Dioxygen

(B) Dihydrogen

(C) Dinitrogen

(D) Ozone

Q.6

By which of the following methods, H_2O_2 can't synthesized?

(A) Aerial oxidation of 2 -ethyl anthraquinol

(B) Addition of ice cold H_2SO_4 on PbO_2

(C) Lewis addition of ice cold H_2SO_4 on BaO_2

(D) Electrolysis of $(NH_4)_2\ SO_4$ at a high current density

Q.7

A green coloured solution of same salt changes its colour to light pink on passing ozone through it. Which of the following species represent pink and green colour respectively?

(A) MnO_4^{2-} and MnO_4

(B) Mn^{2+} and MnO_2

(C) MnO_4^- and ${MnO_4}^{2-}$

(D) Cu^+ and Cu^{2+}

Q.8

Which of the following does not give oxygen on heating?

(A) $K_2Cr_2O_7$

(B) $(NH_4)_2\ Cr_2O_7$

(C) $KClO_3$

(D) $Zn\ (ClO_3)_2$

Q.9

Which of the following on thermal decomposition gives oxygen gas ?

(A) Ag_2O

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(B) Pb_3O_4

(C) PbO_2

(D) All of these

Q.10

Correct order of $O-O$ bond length (increasing) in O_2, H_2O_2 and O_3 is

(A) $H_2O_2 < O_3 < O_2$

(B) $O_2 < O_3 < H_2O_2$

(C) $O_3 < O_2 < H_2O_2$

(D) $O_3 < H_2O_2 < O_2$

Q.11

Which of the following statements regarding ozone is not correct?

(A) The ozone molecule is angular in shape

(B) The ozone is a resonance hybrid of the two structures

(C) The oxygen-oxygen bond length in ozone is identical with that of molecular oxygen

(D) Ozone is used as germicide and disinfectant for the purification of air

Q.12

Which of the following oxidation state of oxygen is not observed in its compounds

(A) $-1/2$

(B) -1

(C) $+4$

(D) $+2$

Q.13

An organic compound containing sulphur is estimated by Carius method in which fuming HNO_3 is used to convert S into :

(A) SO_3^{2-}

(B) SO_4^{2-}

(C) SO_3

(D) SO_2

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Q.14

Which of the following statements about the depletion of ozone layer is correct?

(A) The problem of ozone depletion is less serious at poles because NO_2 solidifies and is not available for consuming ClO' radicals.

(B) The problem of ozone depletion is more serious at poles because ice crystals in the clouds over poles act as catalyst for photochemical reactions involving the decomposition of ozone by Cl^* and ClO° radicals.

(C) Freons, chlorofluorocarbons, are inert chemically, they do not react with ozone in stratosphere.

(D) Oxides of nitrogen also do not react with ozone in stratosphere.

Q.15

Photochemical smog consists of excessive amount of X, in addition to aldehydes, ketones, peroxy-acetyl nitrile (PAN), and so forth, X is:

(A) CH_4

(B) CO

(C) CO_2

(D) O_3

Answers & Solutions

Q.1 Answer:

Acidic oxides

Solution:

Acidic oxides

Q.2 Answer:

K_2MnO_4

Solution:

K_2MnO_4

Q.3 Answer:

Increases to 1.5 times

Solution:

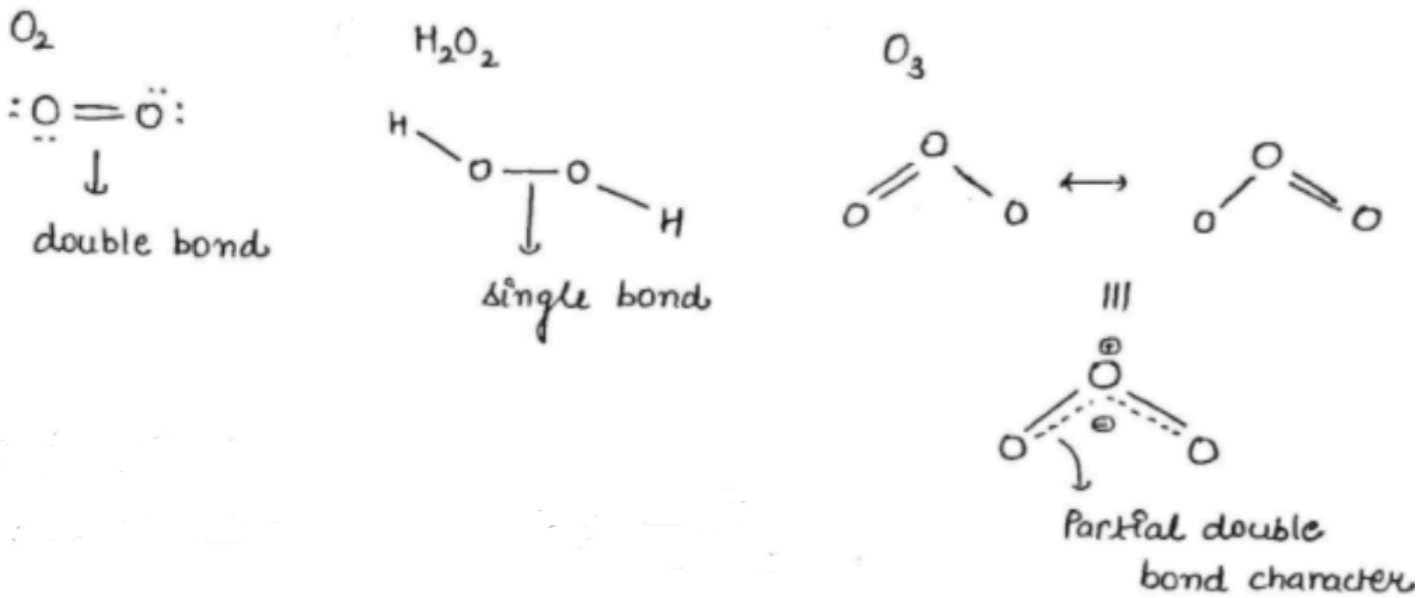
Increases to 1.5 times

Q.4 Answer:

$H_2O_2 > O_3 > O_2$

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Solution:



Bond strength $\rightarrow$

Double bond $>$ **Partial double bond** $>$ **single bond**

Bond length $\rightarrow$

Double bond $<$ Partial double bond $<$ **Single bond**

So, Bond length $(0-0)$ **order is** $\rightarrow$

$H_2O_2 > O_3 > O_2$

Q.5 Answer:

Ozone

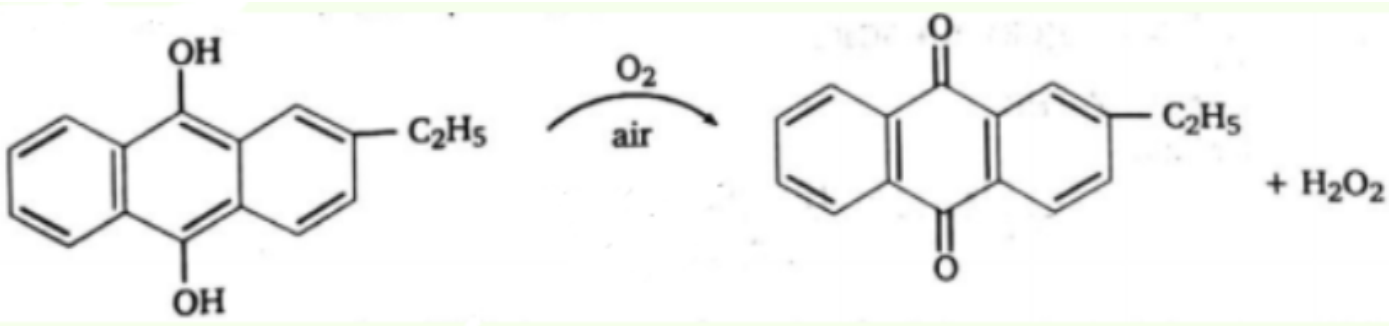
Solution:

When ozone is passed through mercury, it loses its meniscus and sticks to the glass due to the formation of mercurous oxide So, mercury when starts moving on a Surface it leaves a tail and thes is what tailing of mercury is called.

Q.6 Answer:

Addition of ice cold H_2SO_4 on PbO_2

Solution:



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$H_2SO_4 + PbO_2 \rightarrow PbSO_4 \downarrow + \frac{1}{2}O_2 + H_2O$ (PbO_2 **is not peroxy compound)**

$$H_2SO_4 + BaO_2 \rightarrow BaSO_4 \downarrow + H_2O_2$$

$$(NH_4)_2\, SO_4 \leftrightharpoons 2NH_4^+ + SO_4^{2-}$$

At anode : $2SO_4^{2-} \overset{\text{Electrolysis}}{\rightarrow} S_2O_8^{2-} + 2e^-$

Peroxo sulphate on hydrolysis, produces H_2O_2**.**

Q.7 Answer:

MnO_4^- and $MnO_4{}^{2-}$

Solution:

Ozone is a strong oxidizing agent as it has a tendency to readily dewmpose on heating lo give a molecule of oxygen and nascent oxygen,

The green coloured solution can be potassium manganate solution (K_2MnO_4) **having manganate ion** $\left(MnO_4{}^{2-}\right)$**.**

The light pink solution can be Potassium permonganate $(kMnO_4)$ **having permangate ion** $\left(MnO_4^-\right)$**,**

Ozone oxidizes $\dot{M}_n^{+6}O_4^{2-}$ **(green in colour) to** $Mn^{+7}O_4^-$ **(Light pink in colour).**

Q.8 Answer:

$(NH_4)_2\, Cr_2O_7$

Solution:

$$2KClO_3 \rightarrow 2KCl + 3O_2 \uparrow$$

$$2Zn\,(ClO_3)_2 \rightarrow ZnCl_2 + 3O_2 \uparrow$$

$$4\ K_2Cr_2O_7 \rightarrow 4\ K_2CrO_4 + 2Cr_2O_3 + 3O_2 \uparrow$$

$$(NH_4)_2\, Cr_2O_7 \rightarrow Cr_2O_3 + 4H_2O + N_2 \uparrow$$

All the given compounds are oxygen-rich. All of them, except ammonium dichromate

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$[(NH_4)_2\ Cr_2O_7]$, **give oxygen on heating. Ammonium dichromate, on the other hand, gives nitrogen on heating.**

Q.9 Answer:

All of these

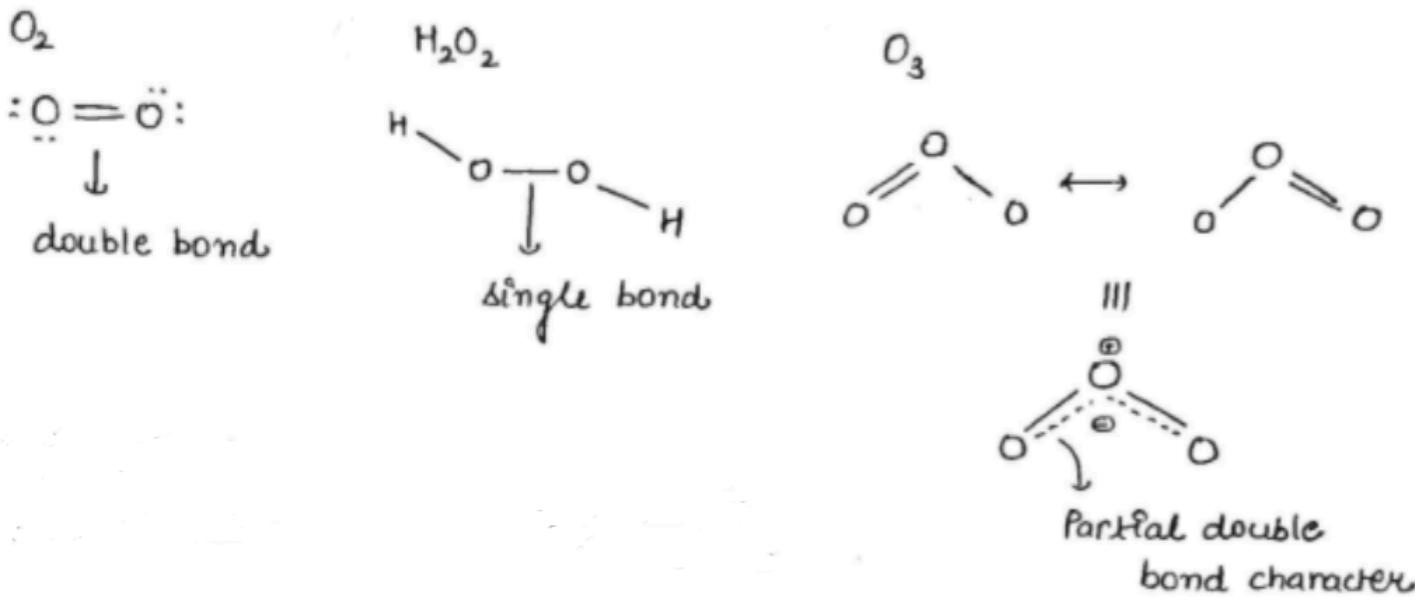
Solution:

$$2Ag_2O(s) \longrightarrow 4Ag(s) + O_2(\ g)$$
$$2\ Pb_3O_4(\ s) \longrightarrow 6PbO(s) + O_2(\ g)$$
$$2PbO_2(\ s) \longrightarrow 2PbO(s) + O_2(\ g)$$

Q.10 Answer:

$O_2 < O_3 < H_2O_2$

Solution:



Bond strength $\rightarrow$

Double bond $>$ **Partial double bond** $>$ **single bond**

Bond length $\rightarrow$

Double bond $<$ Partial double bond $<$ **Single bond**

So, Bond length $(0-0)$ **order is** $\rightarrow$

$O_2 < O_3 < H_2O_2$

Q.11 Answer:

The oxygen-oxygen bond length in ozone is identical with that of molecular oxygen

Solution:

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The bond length in ozone is intermediate between those of single $O-O$ and double $O=O$ bonds. The bond angle is O_3 is 117° with $O-O$ distance $127.8 \mathrm{pm}$.

Q.12 Answer:

$+4$

Solution:

O do not show $+4$ oxidation state because its valency is only 2. $2s$ electrons takes more energy to ionize.

Q.13 Answer:

SO_4^{2-}

Solution:

S is converted to H_2SO_4 $\left(SO_4^{2-}\right)$.

Q.14 Answer:

The problem of ozone depletion is more serious at poles because ice crystals in the clouds over poles act as catalyst for photochemical reactions involving the decomposition of ozone by Cl^* and ClO° radicals.

Solution:

The problem of ozone depletion is more serious at poles because ice crystals in the clouds over poles act as catalyst for photochemical reactions involving the decomposition of ozone by Cl^* and ClO° radicals.

Q.15 Answer:

O_3

Solution:

Photochemical smong also contains O_3.

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